Practice workbook · Areas of regions made of several pieces

Finding the area of a region built from an integral plus or minus a triangle, or from two curves each on its own interval.

6 question parts: 1 worked-example retry and 5 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-CAL-21

1. Question 11 · part c · 7 marks

2023 JAN · Paper 02

Worked-example retry · Attempt part c only.

11

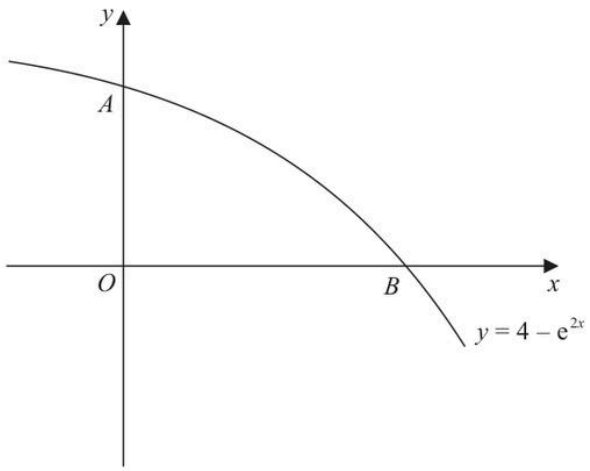


Diagram **NOT** accurately drawn

Figure 3

Figure 3 shows part of the curve C with equation $y = 4 - e^{2x}$
The curve C crosses the y-axis at the point A and the x-axis at the point B.

(a) (i) Write down the y coordinate of point A.

(ii) Show that the x coordinate of B is $x = \ln 2$

(3)

The line l is the normal to C at the point B.

(b) Find an equation for l, giving your answer in the form $y = mx + c$

(4)

The finite region R is bounded by C, l and the y-axis.

(c) Using calculus, find the area of R.
Give your answer to one decimal place.

(7)

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1. Question 11 · part c · 7 marks

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Worked-example retry · Attempt part c only.

Question 11 continued

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1. Question 11 · part c · 7 marks

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Worked-example retry · Attempt part c only.

Question 11 continued

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1. Question 11 · part c · 7 marks

2023 JAN · Paper 02

Worked-example retry · Attempt part c only.

Question 11 continued

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1. Question 11 · part c · 7 marks

2023 JAN · Paper 02

Worked-example retry · Attempt part c only.

Question 11 continued

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(Total for Question 11 is 14 marks)

1. Question 11 · part c · 7 marks · Working and answer

Working and answer

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2. Question 11 · part b · 11 marks

2025 JUN · Paper 02 (regional R)

Further practice · Attempt part b only.

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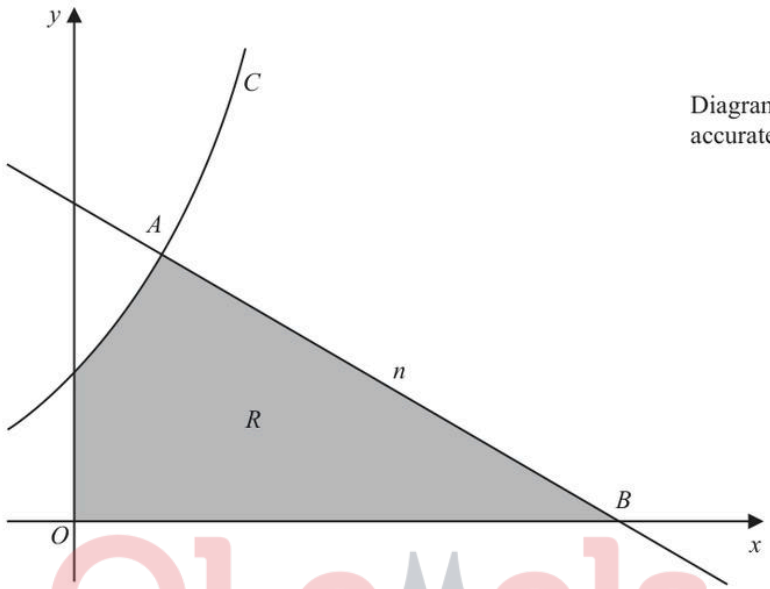


Diagram **NOT** accurately drawn

Figure 4

Figure 4 shows part of a curve C with equation $y = 3e^{\frac{x}{3}}$
The point A with coordinates $(2, p)$ lies on C

(a) Write down the exact value of p

(1)

The straight line n, shown on Figure 4, is the normal to C at A and crosses the x-axis at point B

The region R, shown shaded in Figure 4, is bounded by C, n, the x-axis and the y-axis.

(b) Use algebraic integration to find the area of R

Give your answer in the form $\frac{W}{V}e^2 + We^{\frac{2}{3}} - W$

where W and V are integers to be found.

(11)

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2. Question 11 · part b · 11 marks

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Further practice · Attempt part b only.

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Question 11 continued

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2. Question 11 · part b · 11 marks

2025 JUN · Paper 02 (regional R)

Further practice · Attempt part b only.

Question 11 continued

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(Total for Question 11 is 12 marks)

2. Question 11 · part b · 11 marks · Working and answer

Working and answer

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3. Question 7 · part c · 6 marks

2024 JUN · Paper 01

Further practice · Attempt part c only.

7

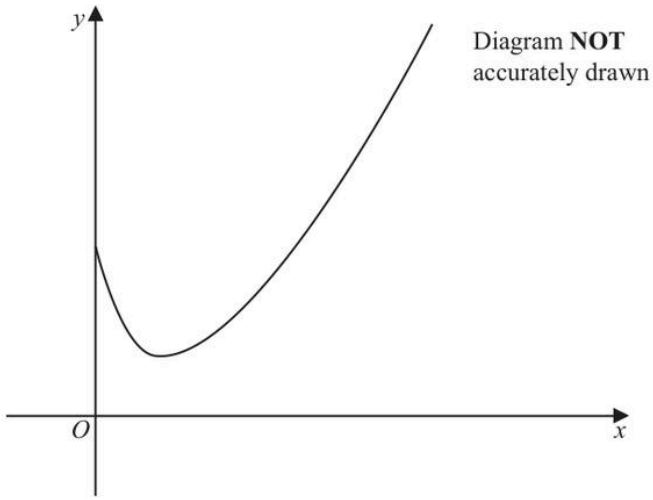


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows a sketch of part of the curve C with equation

$$y = \frac{x^2}{4} - 3\sqrt{x} + 8$$

The point P lies on C and has coordinates $(4, a)$

(a) Show that $a = 6$

(1)

The line L is the normal to C at the point P

(b) Show that an equation of L is $5y + 4x - 46 = 0$

(6)

The finite region R is bounded by the curve C, the line L, the x-axis and the line with equation $x = 1$

(c) Use calculus to find the exact area of R

(6)

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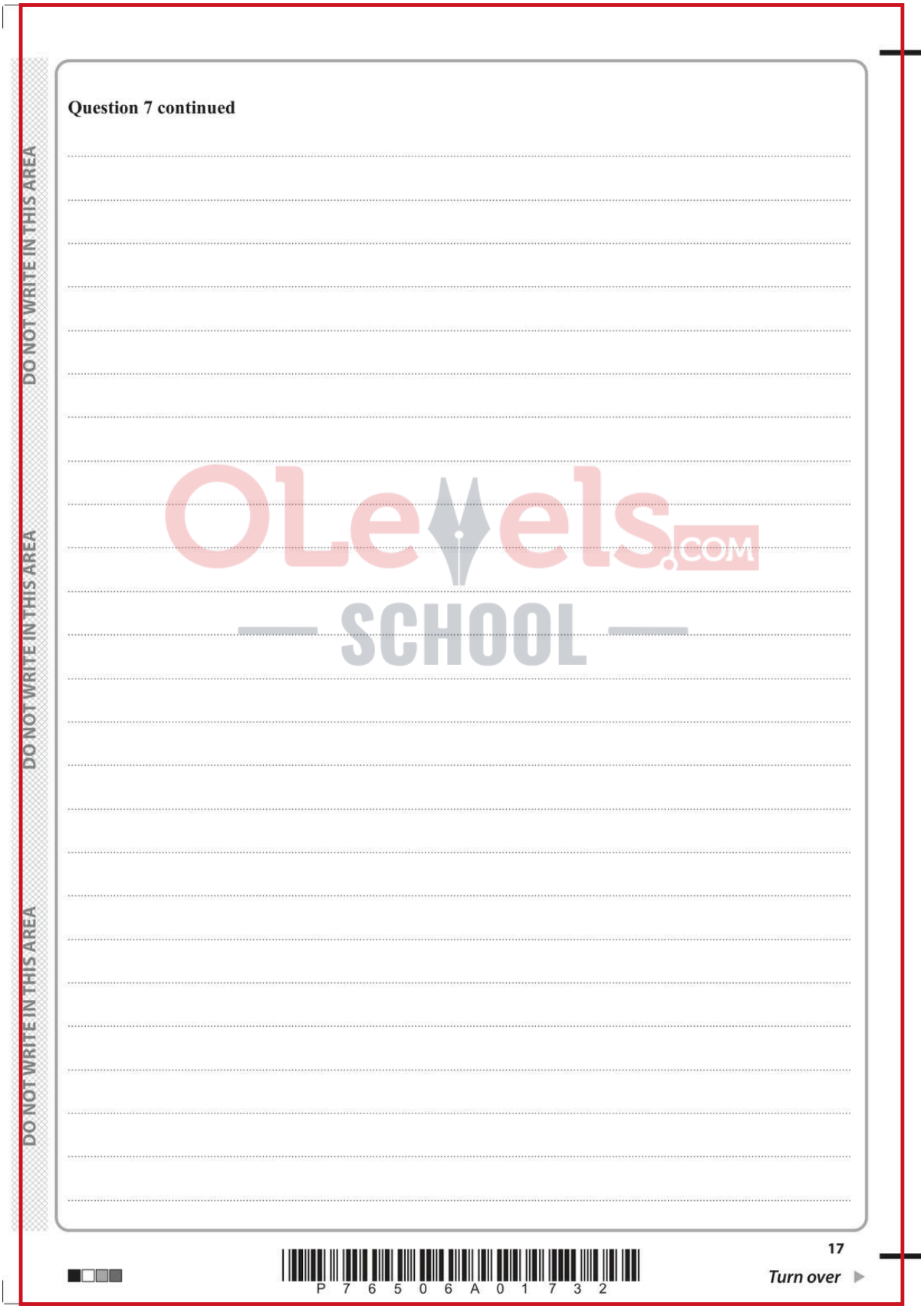
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3. Question 7 · part c · 6 marks

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Further practice · Attempt part c only.



Question 7 continued

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Turn over

3. Question 7 · part c · 6 marks

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Further practice · Attempt part c only.

Question 7 continued

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3. Question 7 · part c · 6 marks

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Further practice · Attempt part c only.

Question 7 continued

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(Total for Question 7 is 13 marks)

3. Question 7 · part c · 6 marks · Working and answer

Working and answer

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4. Question 5 · part b · 7 marks

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Further practice · Attempt part b only.

5

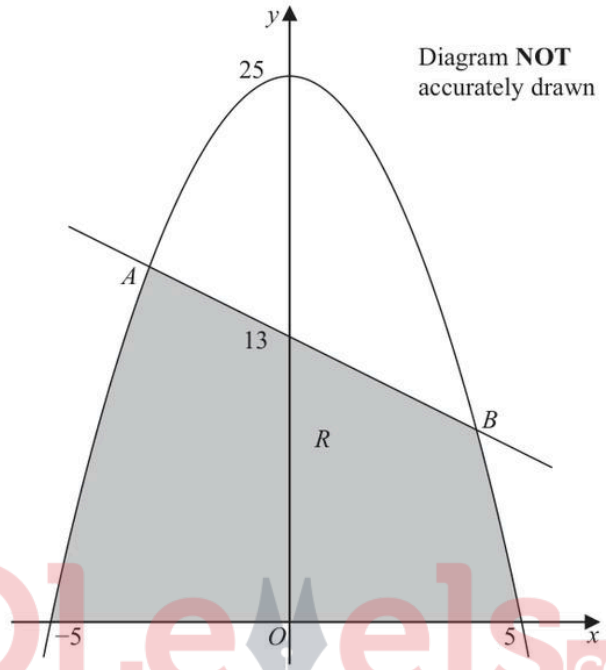


Figure 2

Figure 2 shows part of the curve with equation $y = 25 - x^2$ and part of the line with equation $y + x = 13$

The curve and the line intersect at the points A and B.

(a) Use algebra to find the coordinates of A and the coordinates of B.

(4)

The region R, shown shaded in Figure 2, is bounded by the curve, the straight line and the x-axis.

(b) Use algebraic integration to find the area of R.

(7)

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4. Question 5 · part b · 7 marks

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Question 5 continued

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4. Question 5 · part b · 7 marks

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Further practice · Attempt part b only.

Question 5 continued

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4. Question 5 · part b · 7 marks

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Further practice · Attempt part b only.

Question 5 continued

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(Total for Question 5 is 11 marks)

4. Question 5 · part b · 7 marks · Working and answer

Working and answer

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5. Question 9 · part c · 8 marks

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Further practice · Attempt part c only.

9 (a) Using a formula given on page 2, show that

$$\cos 2\theta = 2\cos^2\theta - 1$$

(2)

(b) Hence show that

$$\int_{\frac{\pi}{3}}^{\frac{3\pi}{4}} \left(2\cos^2\theta - 1\right) \mathrm{d}\theta = -\frac{a+\sqrt{b}}{c}$$

where a, b and c are integers to be found.

(4)

Diagram **NOT** accurately drawn

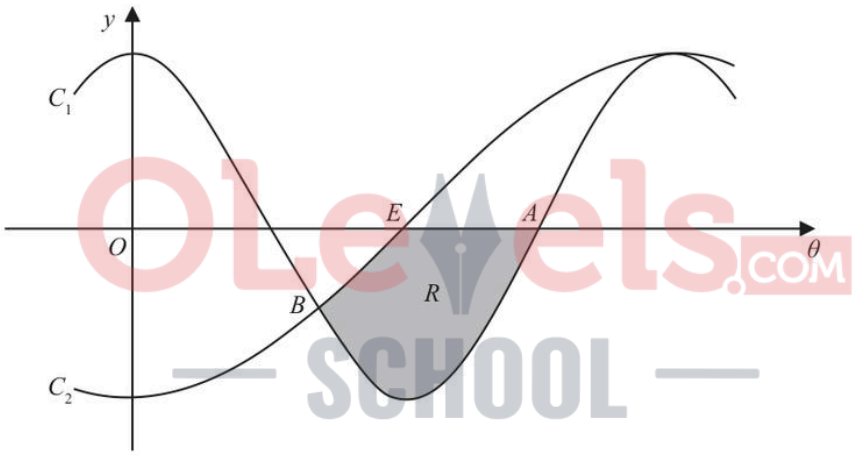


Figure 3

Figure 3 shows part of the curve C_1 with equation $y = 2\cos^2\theta - 1$ and part of the curve C_2 with equation $y = -\cos\theta$

Point B is the intersection of C_1 and C_2 as shown in Figure 3

Point $A\left(\frac{3\pi}{4}, 0\right)$ is the intersection of C_1 with the θ-axis as shown in Figure 3

Point $E\left(\frac{\pi}{2}, 0\right)$ is the intersection of C_2 with the θ-axis as shown in Figure 3

The finite region R, shown shaded in Figure 3, is bounded by the θ-axis, C_1 and C_2

(c) Use calculus to find, in its simplest form, the exact area of R

(8)

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5. Question 9 · part c · 8 marks

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Further practice · Attempt part c only.

Question 9 continued

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5. Question 9 · part c · 8 marks

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Further practice · Attempt part c only.

Question 9 continued

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5. Question 9 · part c · 8 marks

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Further practice · Attempt part c only.

Question 9 continued

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(Total for Question 9 is 14 marks)

5. Question 9 · part c · 8 marks · Working and answer

Working and answer

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6. Question 12 · part c · 4 marks

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Further practice · Attempt part c only.

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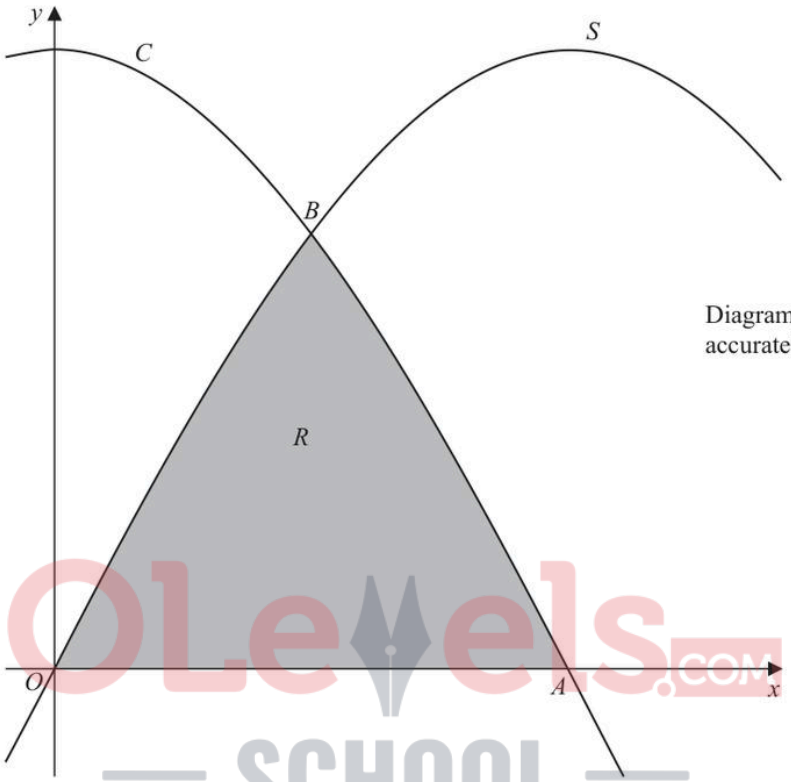


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Figure 2

The region R, shown shaded in Figure 2, is bounded by the x-axis, the curve S with equation $y = 2\sin x$ and the curve C with equation $y = 2\cos x$.
As shown in Figure 2, C crosses the x-axis at the point A.

(a) Write down the x coordinate of A.

(1)

As shown in Figure 2, C and S intersect at the point B.

(b) Find the x coordinate of B.

(2)

(c) Using calculus, find the area of the shaded region R.
Give your answer in the form $a - \sqrt{b}$ where a and b are integers.

(4)

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6. Question 12 · part c · 4 marks

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Further practice · Attempt part c only.

Question 12 continued

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6. Question 12 · part c · 4 marks

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Further practice · Attempt part c only.

Question 12 continued

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(Total for Question 12 is 7 marks)

6. Question 12 · part c · 4 marks · Working and answer

Working and answer

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